

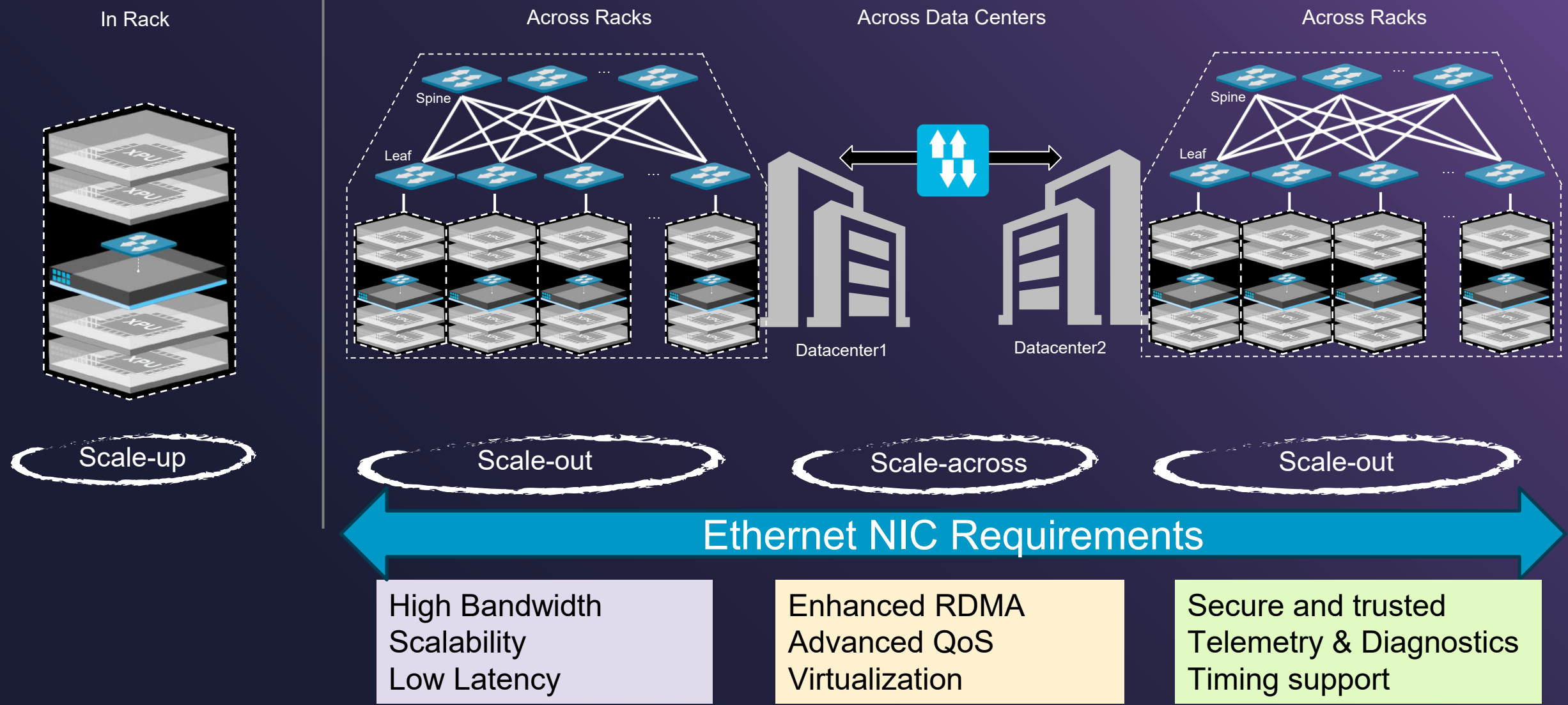


# Thor Ultra Ethernet NIC for AI and HPC Markets

Hemal Shah, Distinguished Engineer and Architect, Broadcom Inc.

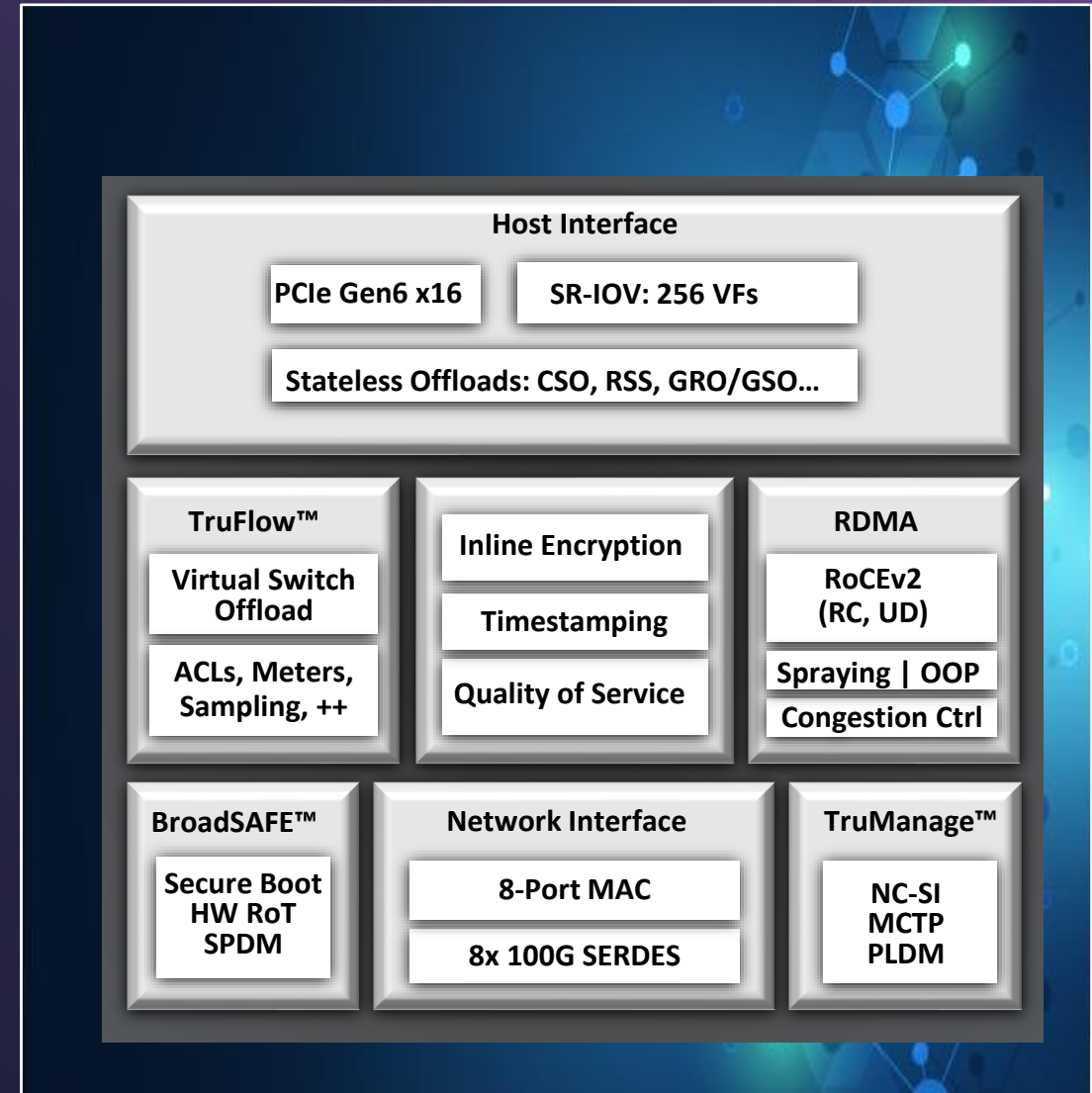


# Ethernet Networking for AI and HPC

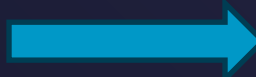


# Introducing Thor Ultra Ethernet NIC

- PCIe Gen6 x16 Host Interface
- 8x 100G PAM4/NRZ Serdes
- Enhanced RoCE with 64K+ QPs
- Hierarchical QoS & SR-IOV
- Timing, Security & Manageability Support



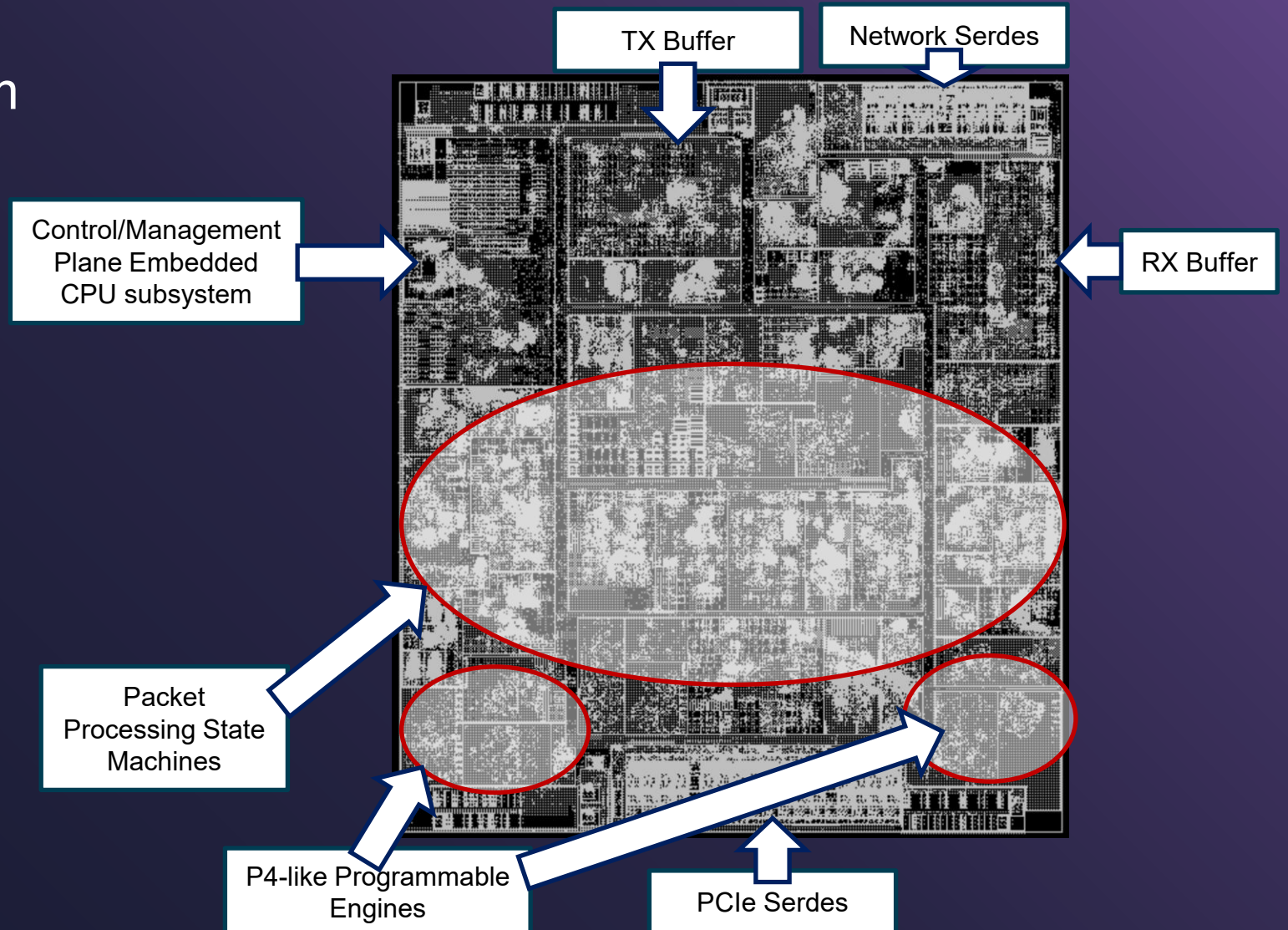
# Thor Ultra Ethernet NIC – Generational Context

- No multipathing  Packet spray & OOO placement
- Inefficient Go-Back N  Reliability enhancements
- Hard to tune congestion control (CC)  Simple & programmable CC
- Port count  2x multiplane connectivity



# Thor Ultra Chip - Physical Design

- Process technology: 5nm
- Transistor count: 2.4B+
- Package: 27x27
- Max Power: 40-42W



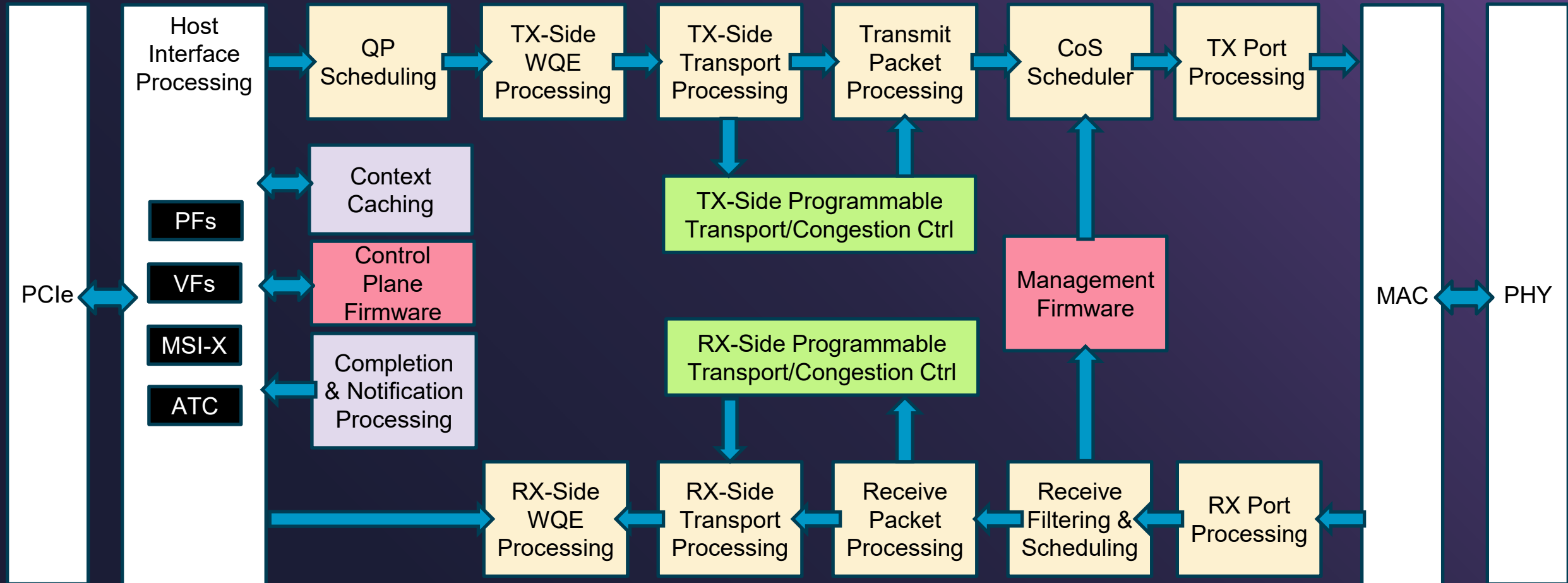
# Thor Ultra NIC Boards



| Description                                 | PCle                                    | OCP NIC 3.0 |
|---------------------------------------------|-----------------------------------------|-------------|
| Form Factor                                 | CEM HHHL                                | TSFF        |
| Connector Configurations                    | 1x OSFP112 and 2x QSFP112               |             |
| Ethernet Port Configurations for OSFP112    | 1x 800G, 2x 400G, 4x 200G, 8x 100G      |             |
| Ethernet Port Configurations for 2x QSFP112 | 2x 400G, 4x (2*2x) 200G, 8x (2*4x) 100G |             |
| Cable/Module Support                        | DAC, LPO, AEC, and Optics               |             |
| Cooling Options                             | Air flow and liquid cooling             |             |
| Max Board Power (w/o optics)                | 50-55W                                  |             |



# Thor Ultra NIC Pipeline



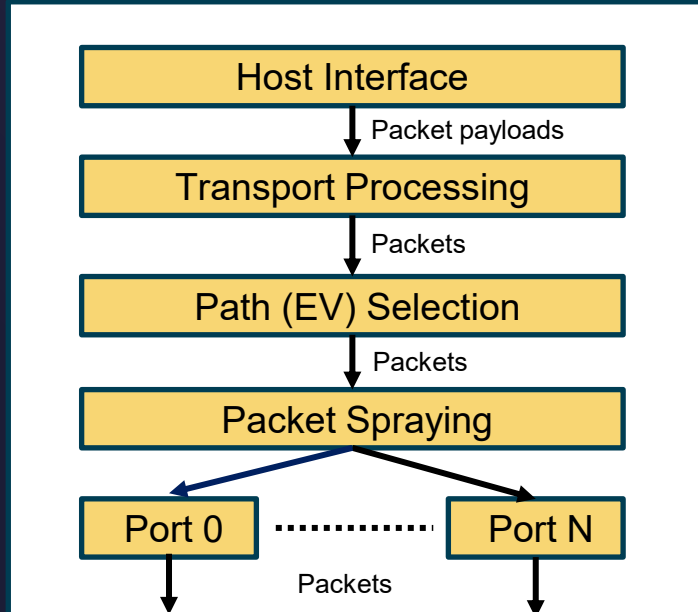
# Thor Ultra enhanced RoCE (eRoCE) Features

| Feature                                | Description                                                                                                                                                                                                                                                         |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Multipath</b>                       | <ul style="list-style-type: none"><li>• Header entropy variation per QP or per packet</li><li>• Switch load balancing modes supported: ECMP, E-ECMP, DLB flowlet, DLB spray</li><li>• NIC packet spraying across multiple ports (planes) – up to 8 planes</li></ul> |
| <b>Out-Of-Order (OOO) Placement</b>    | <ul style="list-style-type: none"><li>• OOO placement of RDMA Write &amp; Read responses (Write w/ Imm delivered in-order)</li><li>• In-order placement/delivery of Sends, Read Requests, and Atomics</li></ul>                                                     |
| <b>Reliable Delivery</b>               | <ul style="list-style-type: none"><li>• Selective Acknowledgement (SACK) &amp; Negative Acknowledgement (NACK)</li><li>• Selective retransmissions</li><li>• Probes to evaluate the quality of a path</li></ul>                                                     |
| <b>Congestion Control Enhancements</b> | <ul style="list-style-type: none"><li>• Receiver Credit Based Congestion Control (RCCC) - baseline</li><li>• Programmable Congestion Control</li></ul>                                                                                                              |
| <b>Telemetry</b>                       | <ul style="list-style-type: none"><li>• ECN for deterministic or probabilistic marking</li><li>• Packet trimming for drop notifications</li><li>• CSIG</li></ul>                                                                                                    |

eRoCE = Multipath Reliable Connection (MRC) ++

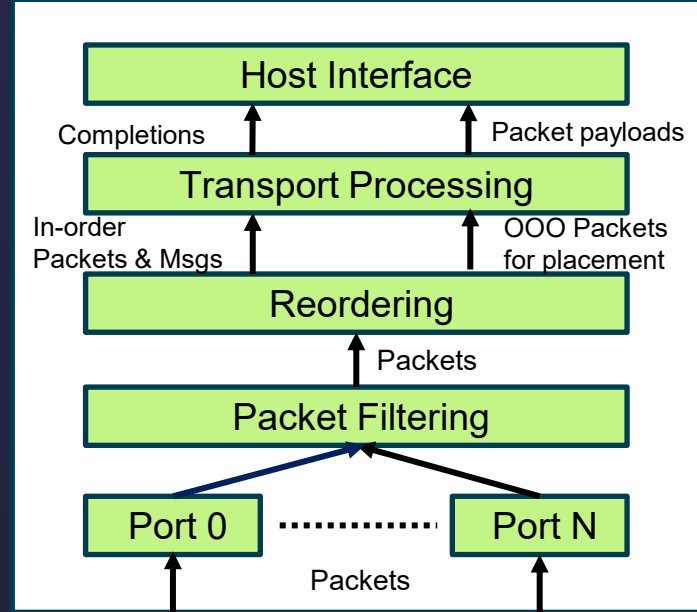


# Thor Ultra eRoCE Implementation



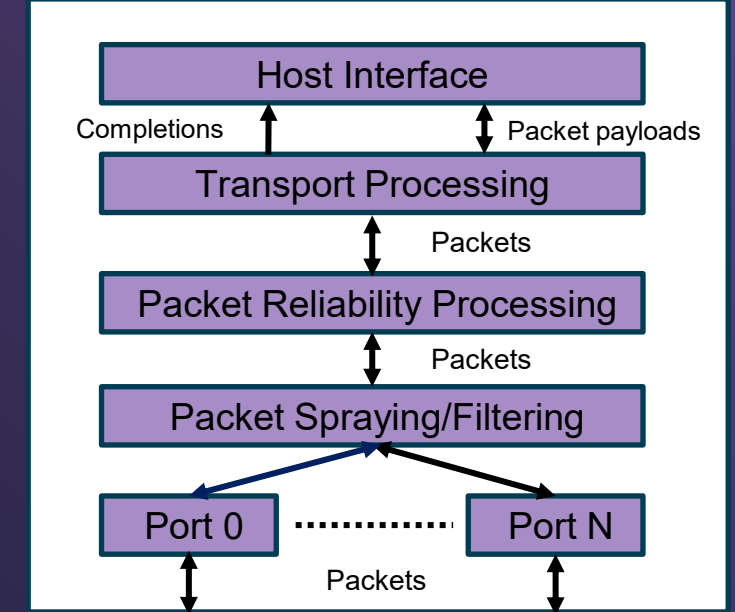
- Path selection per packet
- Paths tracked per QP
- Packets sprayed across ports
- Multiple load balancing modes

## Multipath



- Ext. headers for placement info
- Re-ordering logic at the receiver
- Bitmaps to track received pkts
- In-order message delivery

## OOO Placement



- SACK bitmaps per QP
- Configurable SACK policies
- Packet timers with exp backoff
- NACK for congestion signaling

## Reliable Delivery



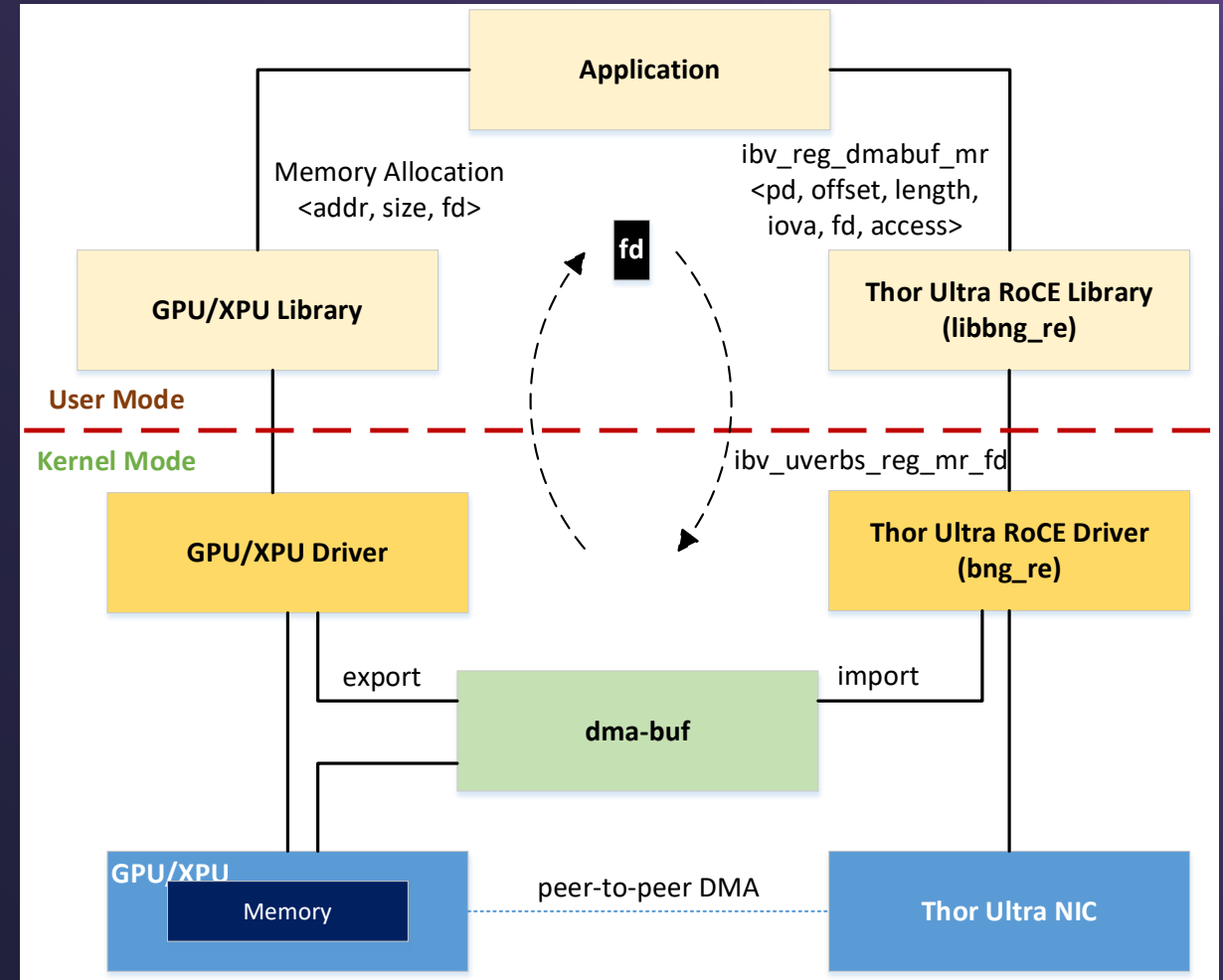
# Thor Ultra Congestion Control Implementation

- Receiver Credit-based Congestion Control (RCCC) – baseline for eRoCE
  - Receiver managed credit allocation to active senders
  - Speculative credits for line rate startup
  - Credit scheduling is dynamically adjusted at the receiver
  - Telemetry (ECN, trimming, and CSIG) aware
- Programmability
  - Match-action (P4-like) programmable engines
  - Enables user defined and/or customizable algorithms



# Thor Ultra NIC Peer Memory Data Transfer Support

- eRoCE for remote data transfers
  - Out-of-order placement of payload
  - In-order delivery of messages
- dma-buf for local peer data transfers

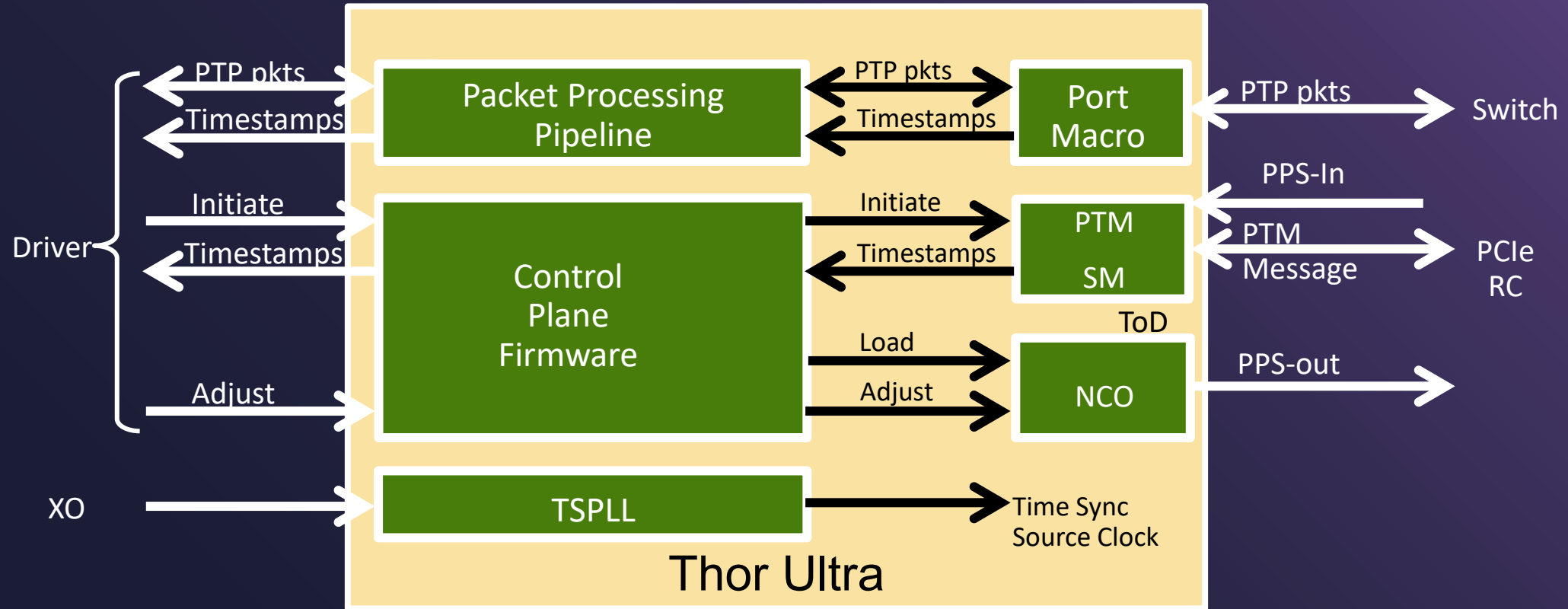


# Thor Ultra Multi-tenancy Support

- QoS
  - Hierarchical scheduling – QP & CoS
  - Traffic shaping in the hardware
  - Rate limits – QP, PCIe function, port
  - Flow level meters with drop
- Virtualization
  - SR-IOV
  - Function level isolation
  - NIC resource virtualization
  - Address Translation Service (ATS)



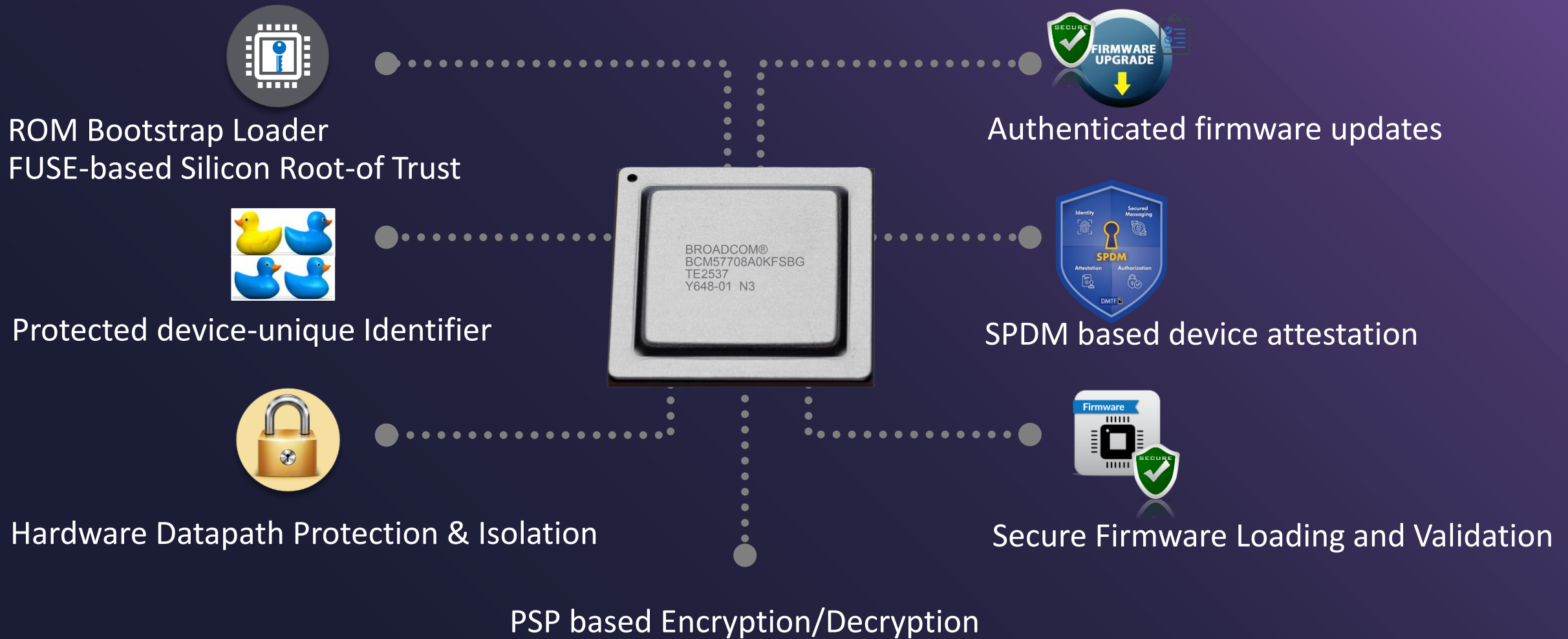
# Thor Ultra Timing Support



- Precision Time Protocol (PTP) v2
- Precision Time Measurement (PTM)
- Packet timestamping
- Time & Freq PHC adjustment
- Time accuracy in ns
- External TCXO



# Thor Ultra NIC Security



# Thor Ultra NIC: Enabling Stability, Control and Visibility

## Health Monitoring



**Continuous health monitoring**

**Chip/Optics temp monitoring**

**Statistics and error reporting**

**Monitor NIC resource utilization**

## Diagnostics



**Logging of adapter states**

**Crash and core dumps**

**Eye diagrams**

**Self diagnostics tests**

## NIC Control



**Gain insights into NIC config**

**Link, Media, and Pre-boot settings**

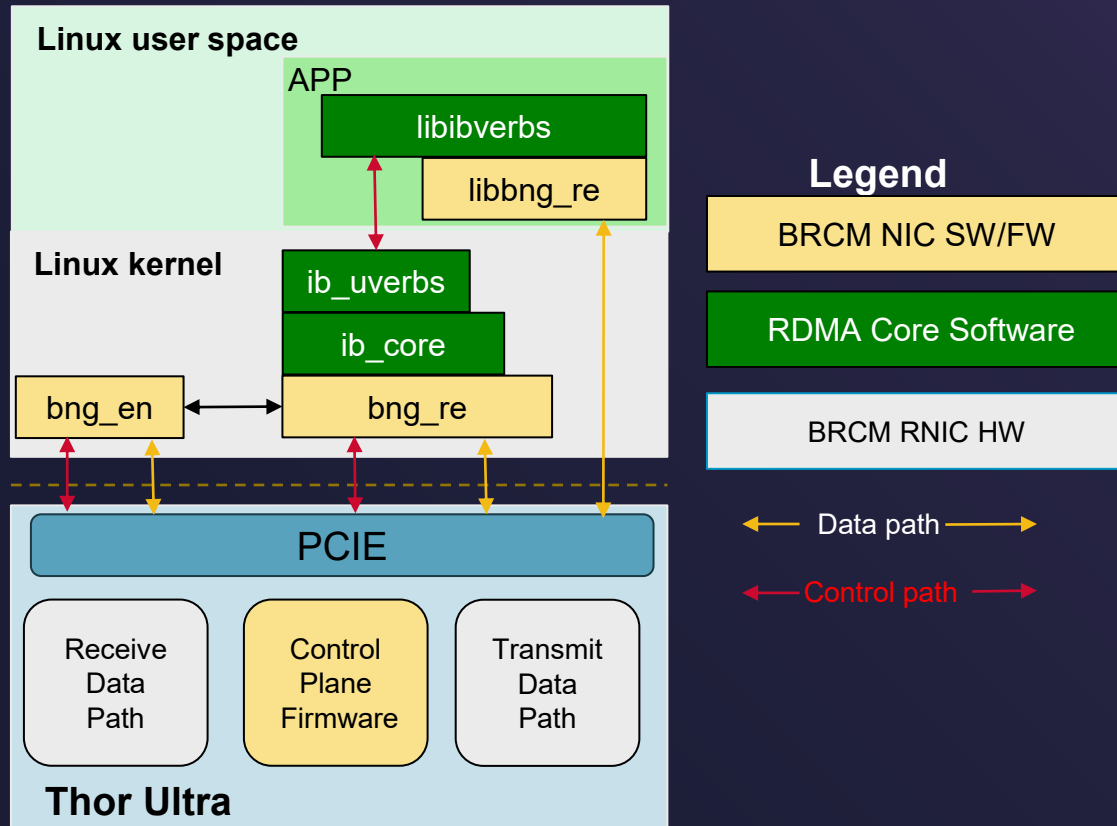
**Authentication & attestation**

**Firmware inventory & update**

**DMTF and OCP Standards Based Manageability**



# RoCE Software Components for Linux OS

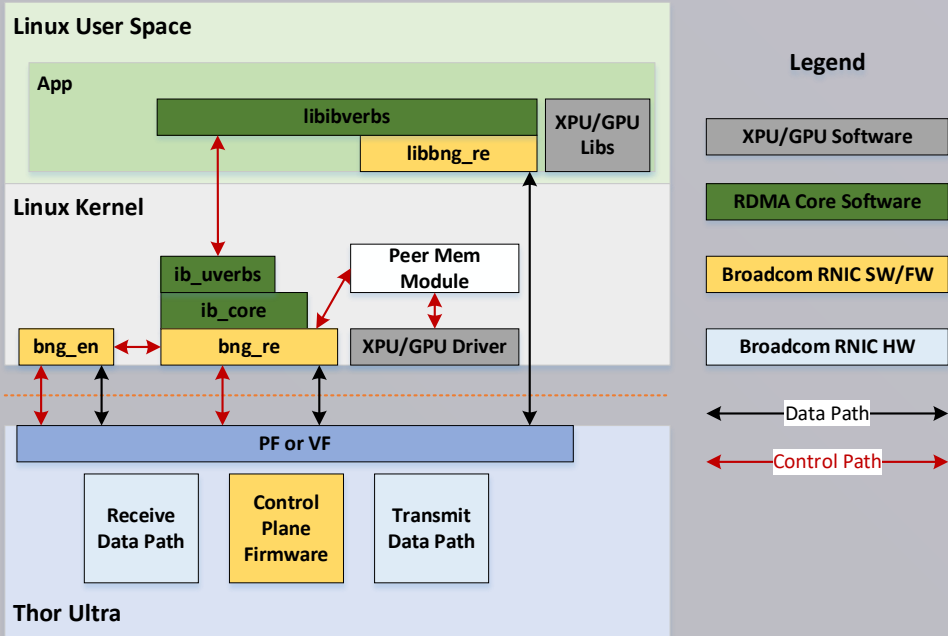


- **Broadcom Linux RoCE Components**
  - RoCE User Library (libbng\_re)
  - RoCE driver (bng\_re)
  - NIC driver (bng\_en)
- **RoCE Driver (bng\_re)**
  - Netdev Event based interface with L2
  - Kernel mode verbs support for RC and UD
  - Kernel mode direct verbs for eRoCE
- **RoCE User library (libbng\_re)**
  - User mode verbs support for RC and UD
  - Direct verbs APIs for eRoCE



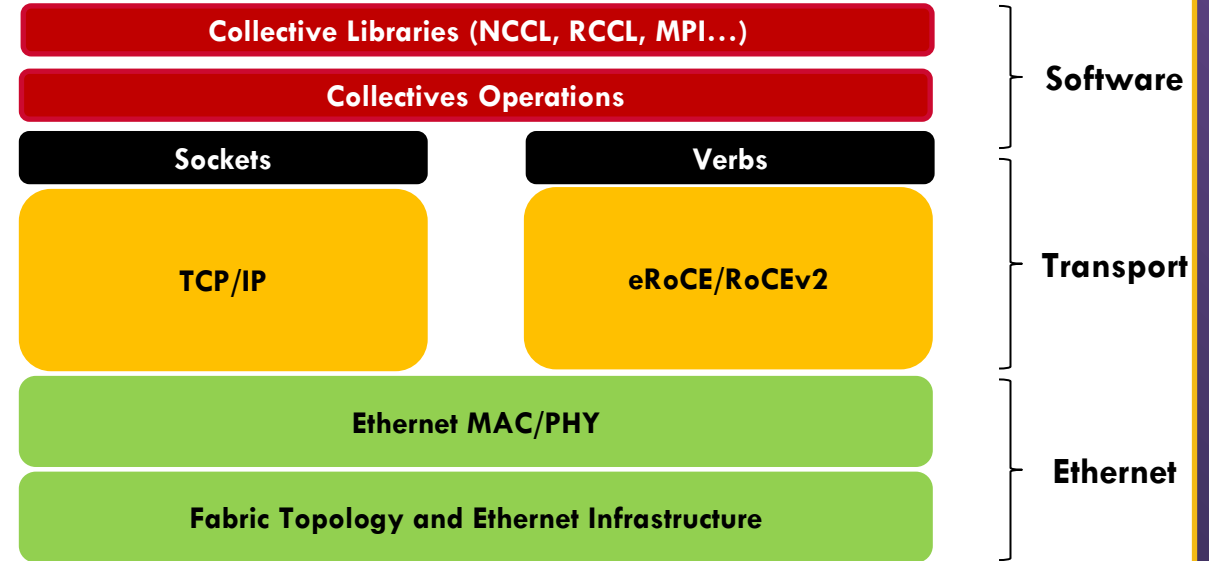
# XPU/GPU Infrastructure Support

## Peer Memory Direct



- Peer memory model
- Linux dma-buf based (upstream)
- XPU/GPU agnostic

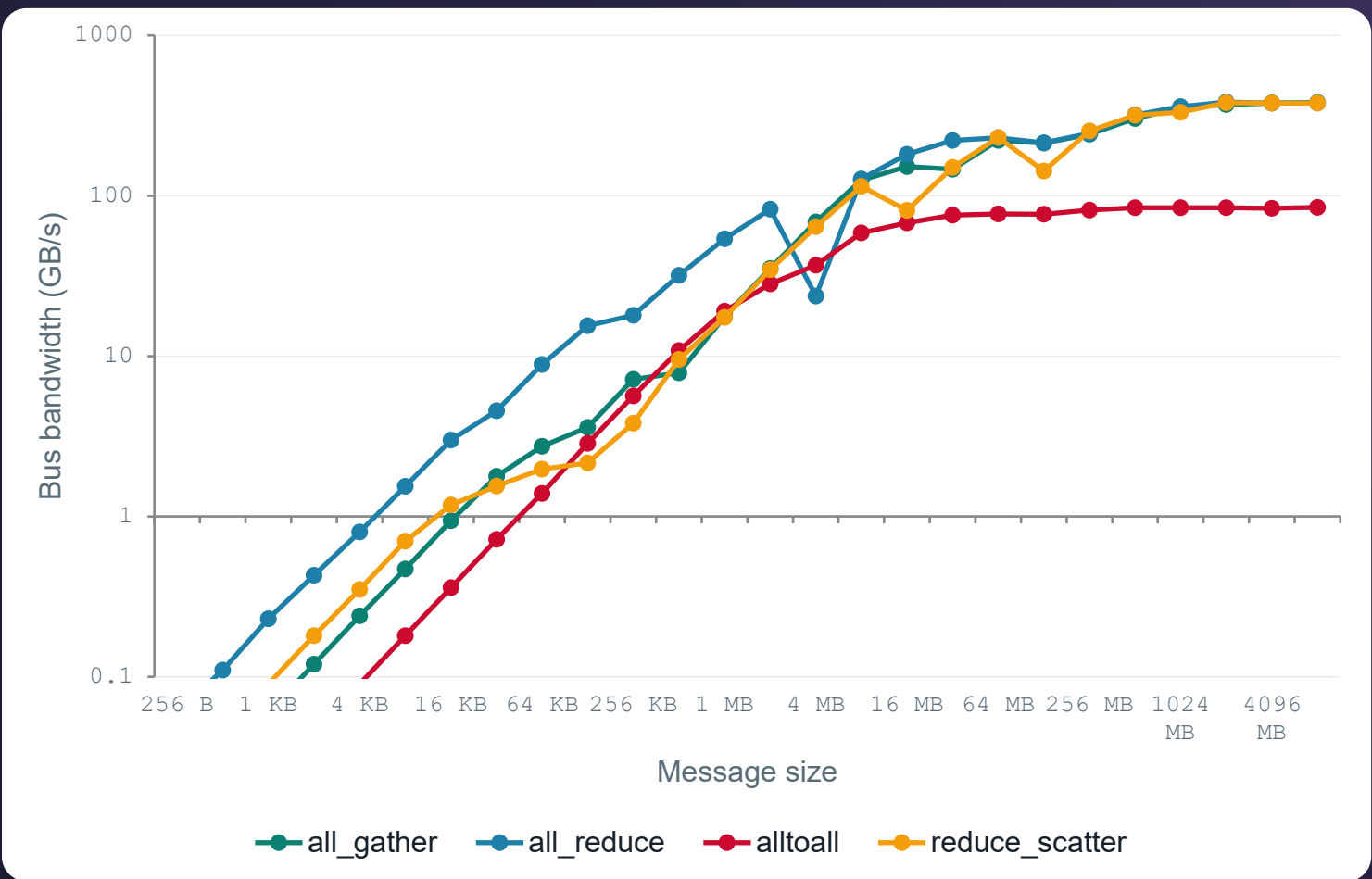
## Collectives



- Enabled by Verbs Provider (upstream) & xCCL Plug-ins
- Unmodified applications
- Multiple Collective Libraries supported
  - xCCL, NCCL, RCCL, MPI....



# Collective Performance – Initial Data on a Gen5 GPU Platform



| Collective     | Peak GB/s | @ size | vs ceiling |
|----------------|-----------|--------|------------|
| all_gather     | 381.92    | 8GB    | 400 GB/s   |
| all_reduce     | 383.93    | 2GB    | 400 GB/s   |
| alltoall       | 84.62     | 8GB    | 93 GB/s    |
| reduce_scatter | 380.23    | 2GB    | 400 GB/s   |

- 2 nodes
- 8 Gen5 GPUs per node
- 16 ranks
- 16 x 400G connectivity to Switch

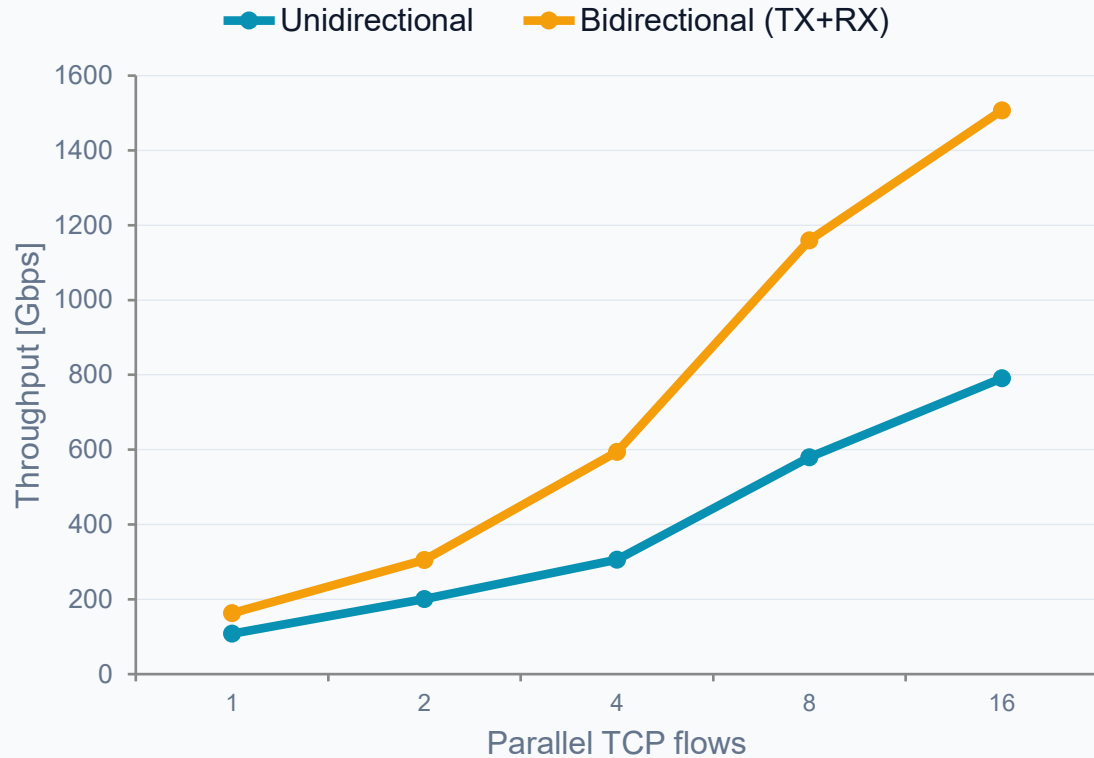


## Near Line-rate

Bus bandwidth > 96% — for gather, reduce, reduce\_scatter



# Thor Ultra TCP Throughput



CPU / memory  
OS / kernel  
NIC driver  
TCP test

**Gen6 CPU · 512 GB DDR5-8000**  
**Ubuntu 24.04.1 · 6.17**  
**1.140.1.85**  
**iperf3 3.21+ (tuned buffers)**



## Near line-rate TCP

Unidirectional iperf3 hits 791 Gbps at 16 flows — 98.9% of the 800G link



## Full-duplex headroom

Bidirectional aggregates to ~1.51 Tbps; TX and RX each sustain ~750 Gbps

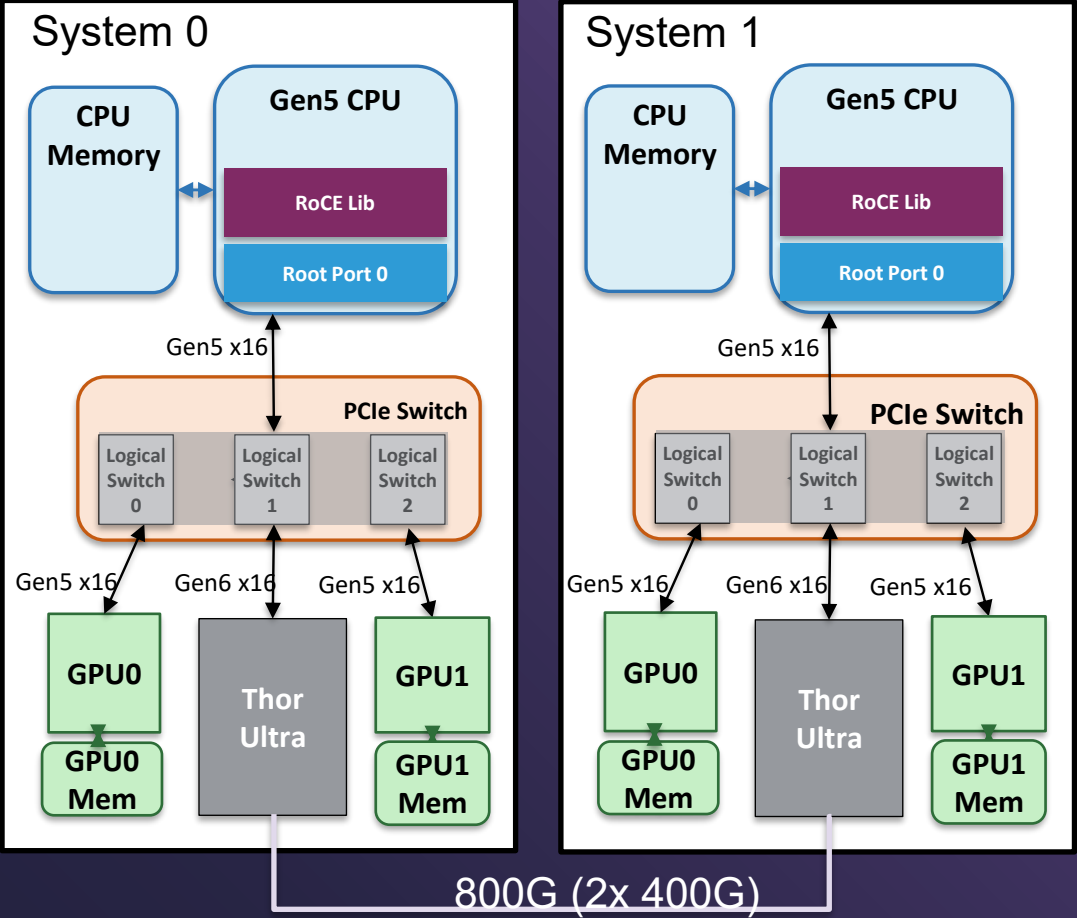
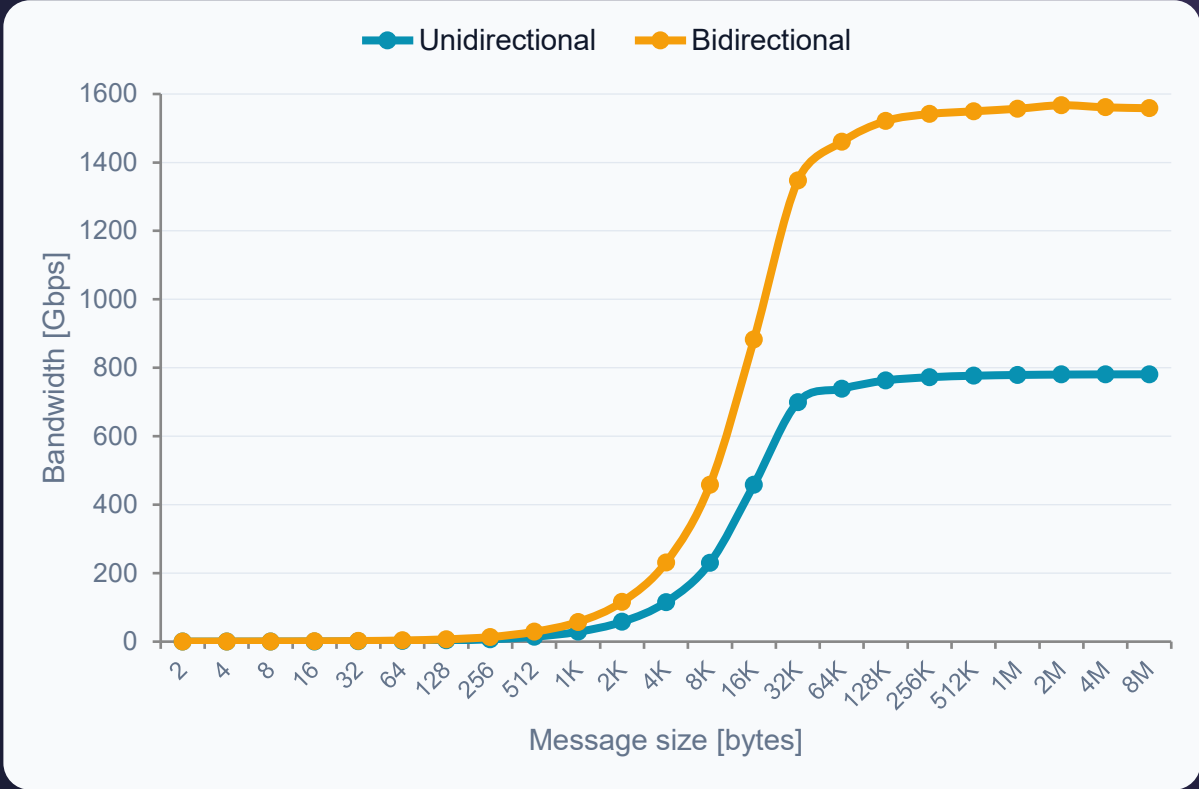


## Scales with flows

Throughput climbs near-linearly to 8 flows, then tapers as the link saturates.



# Thor Ultra Throughput for RDMA Writes



## Line-rate RDMA

Unidirectional saturates at 781 Gbps — 97.6% of 800G — for large messages.



## Full-duplex $\approx 2X$

Bidirectional reaches 1558 Gbps (97.9% of 1.6 Tb/s), roughly double unidirectional.



## Fast ramp

Unidirectional reaches  $\sim 88\%$  of line rate by 32 KB and within  $\sim 3\%$  of peak by 128 KB.



# Hard Engineering Problems Solved

- Multipathing → EV state machines and tracking in packet processing pipeline
- RoCE enhancements → reliability, OOO placement/re-order logic, pkt timers...
- Programmable congestion control → P4-like programmability in the pipeline



# Summary

- Thor Ultra  800G NIC for AI/HPC markets
- Thor Ultra data path  designed to support enhanced RoCE (MRC++)
- QoS and Virtualization features  enable XPU/CPU as a service
- Industry leading security and manageability  easy to use and trusted
- Timing features  enable synchronization of AI/HPC systems





# Thor Ultra Linux Tools

